TSA 2010

Problem Solving

Questions taken from the TSA Oxford 2010 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2010 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) Splashford Swimming Pool charges £2 per session for adults and £1 for children.

Also available is a Family Swimcard. At a cost of £50, the Family Swimcard allows unlimited use of the pool for one year for 2 adults and up to 3 children. For larger families, every additional child must pay half the children's rate each time.

Mr and Mrs Teal and their 4 children are keen swimmers. They used their Swimcard when the family went swimming 40 times last year.

How much did the Swimcard save the Teal family last year?

A £ 50
B £ 190
C £ 230
D £ 250
E £ 270

2 (Q6) Acme Haulage pays dividends to its shareholders every six months. Better Ball Bearings pays its shareholders every eight months and Koffmore Chemists pays every eighteen months.

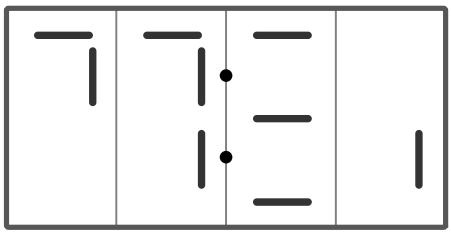
In January 1990 all three companies paid me a dividend.

In what month and year did this next happen?

A July 1991
B January 1992
C July 1995
D January 1996
E January 2002

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3 (Q7) A woman has to get up for work at seven o'clock in the morning. She wakes up and looks over at her digital clock. Unfortunately, the batteries are low and the figures on the display, showing hours and minutes, aren't showing up properly. All she can see is the display below:



It is light outside, so she cannot be more than twelve hours late.

What is the greatest amount of time that she could be late waking up?

A 1h 39min

B 1h 59min

C 2h 25min

D 2h 35min

E 2h 59min

4 (Q8) The Richter Scale is used to measure the strength of earthquakes. It has the property that an increase of one unit on the scale represents a tenfold increase in the strength of the quake. Three earthquakes have the following values on the Richter Scale:

X : 2.0
Y : 3.0
Z : 4.0

Which one of the following statements about their relative strengths is true?

A X is one-tenth the strength of Y and one-twentieth the strength of Z.

B X is one-tenth the strength of Y and one-hundredth the strength of Z.

C X is one-tenth the strength of Y and one-thousandth the strength of Z.

D Z is twice as strong as X and Y is one-and-a-half times as strong as X.

E Z is one-tenth stronger than Y which is one-tenth stronger than X.

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5 (Q12) Sue and Ben are buying some kitchen goods at a shop which is having a sale. The terms of the sale are as follows:

Total marked price of purchases (before discount)	*Reduction (%)*
Up to £50	25
Up to £100	33
£100 or more	50

The marked prices of the goods they have chosen come to £96.

How much less would they actually spend, to the nearest £1, if they chose extra goods to the value of £6?

A £ 2
B £ 6
C £ 10
D £ 13
E £ 16

6 (Q13)

QUARTERLY ELECTRICITY BILL					
Meter Readings		*Units Charged*	*Tariff*	*Pence per unit*	*Amount (£)*
Previous	*Present*				
45856	47846	1990	standard	8.45	168.15
			FIXED CHARGES		12.30
65026	65284	258	Off-peak	3.90	10.06
			FIXED CHARGES		5.45
			TOTAL AMOUNT DUE		195.96

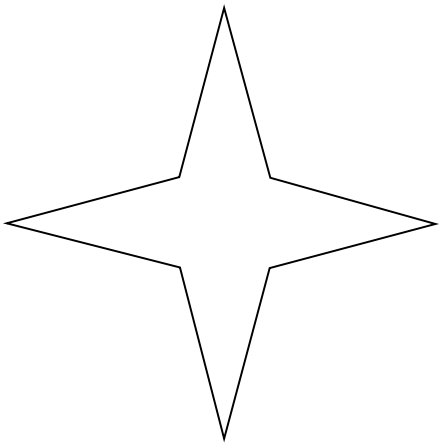
At the end of the quarterly charging period to which the above bill relates, the householder changed her electricity consumption pattern. In the next quarter she roughly halved her standard consumption and increased her off-peak consumption by about ten times.

What approximate total bill should she expect for the next charging period?

A £ 90
B £ 120
C £ 150
D £ 170
E £ 200

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7 (Q14) The diagram shows the outline of a window which is intended to be composed of a number of panes of glass, all of the same size and shape.

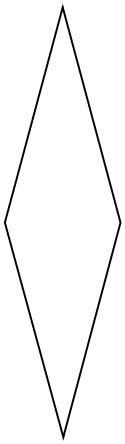


Which one of the shapes below could be used to fill this window?

A



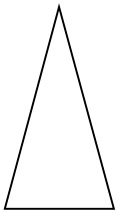
B



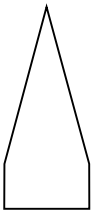
C



D



E



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8 (Q18) I have been asked to move 120 boxes weighing 25kg each and 90 boxes weighing 20kg each from the ground floor to the fifth floor of the office block in which I work. Unfortunately, nobody is available to give me any help.

There is a sign in the lift which reads:

MAXIMUM LOAD 4 persons or 400 kg

I weigh 80kg.

What is the minimum number of journeys I will have to make from the ground floor to the fifth floor?

A 10
B 12
C 13
D 15
E 16

9 (Q19) It is 12 noon and I have just arrived in the centre of town for a meeting that is due to last for at least three and a half hours.

After this meeting, I have an important appointment at 6.00pm which requires me to leave before 5.30pm.

There are 5 car parks in the town centre that charge as follows:

Grove Street	£6.00 per day, flat rate.
Victoria Square	£1.00 per hour or part of an hour.
Central Park	Free for 2 hours, then £2.00 per hour or part of an hour.
Bonningtons	Free for 1 hour, then £1.50 per hour or part of an hour.
Grange Road	£2.50 for 3 hours, then £1.50 per hour or part of an hour.

In which car park should I leave my car in order to pay as little as possible in parking fees?

A Grove Street
B Victoria Square
C Central Park
D Bonningtons
E Grange Road

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10 (Q20) A taxi service advertises its rates for travel to the local airport as shown below:

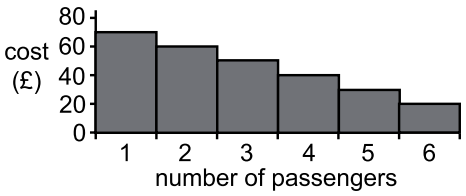
AIRPORT TAXI SERVICE

£70
(includes return journey)

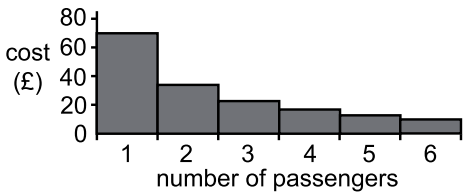
EXTRA PASSENGERS £10 EACH
(vehicles licensed to carry 6 passengers)

Which one of the following bar graphs correctly illustrates how the cost per person decreases as more people get together to share a taxi to the airport?

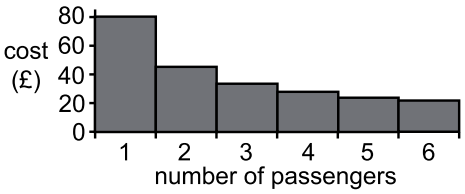
A



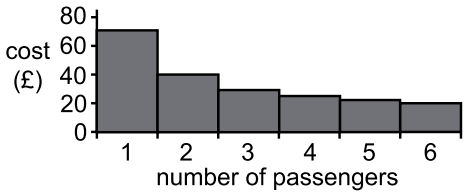
B



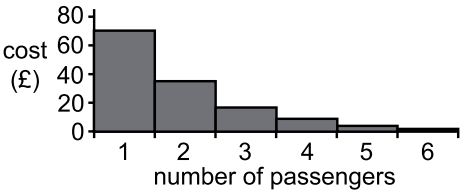
C



D



E



11 (Q24) A secretary is in a bad mood with her boss and has to write three letters intended for three different people. She makes sure that each of the recipients of the letters will receive one written to someone else.

In how many different ways can she send three different letters to achieve this?

A 1
B 2
C 3
D 5
E 6

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12 (Q25) Anne is a researcher working for an insurance company which has decided to send out a safety awareness mailshot to its customers during the month which costs it most in claims. She examines the average accident statistics bearing in mind that an accident categorised as a write-off costs 10 times more than one categorised as serious which, in turn, costs 10 times more than one categorised as slight.

Month	*Write-off*	*Serious*	*Slight*	*Totals*
January	30	539	2891	3460
February	24	414	2513	2951
March	31	550	2972	3553
April	19	401	2901	3321
May	19	440	2993	3452
June	24	509	3230	3763
July	24	515	2899	3438
August	26	477	2732	3235
September	22	510	2867	3399
October	30	474	3236	3740
November	21	578	2895	3494
December	26	381	2967	3374

In which month does Anne conclude the mailshot should be made?

A January

B March

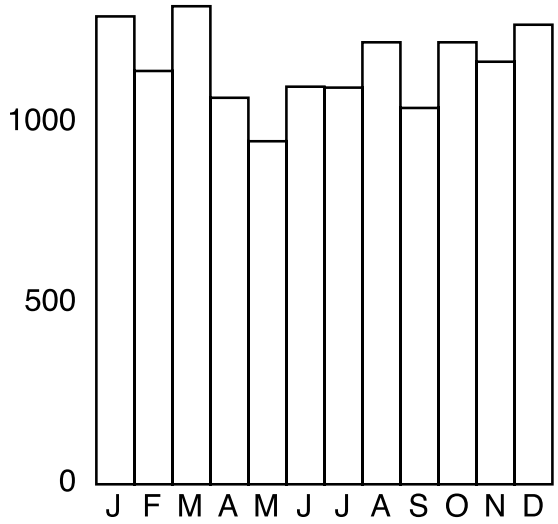
C June

D October

E November

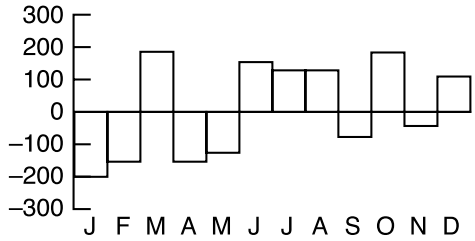
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13 (Q26) The graph below shows Mr Evans' bank balance at the end of each month in a year.

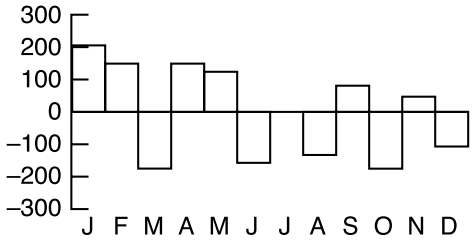


Which one of the following graphs could show the actual change in his bank balance each month?

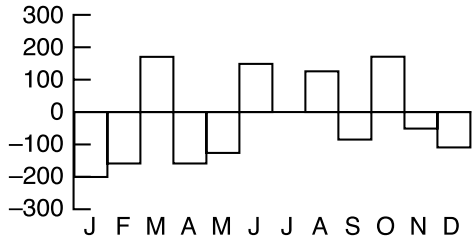
A



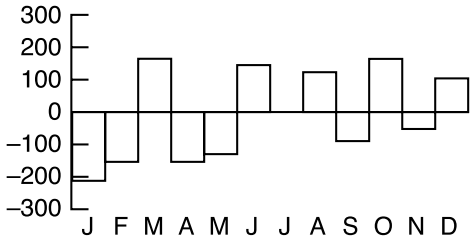
B



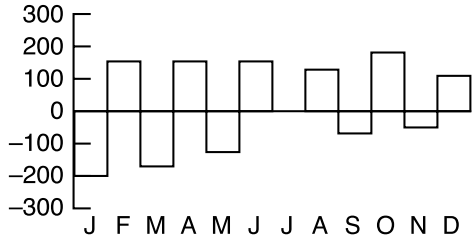
C



D



E



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14 (Q30) Fred, Joe and Bill do plumbing, electrics and plastering respectively and work together on certain building jobs. They will each take on any job involving all three types of work, sub-contract to the others where necessary and sort out any payments at the end of the month.

At the end of March, Bill has done £300 of work for Fred and £200 of work for Joe. Joe has done £100 of work for Fred and £250 of work for Bill. Fred has done £150 of work for Joe and £200 of work for Bill.

Which of the following will settle their accounts?

A Bill and Joe each pay Fred £25.

B Fred pays Bill £50.

C Fred and Joe each pay Bill £75.

D Bill pays Fred and Joe £100 each.

E Joe pays Bill £150 and Fred £50.

15 (Q31) Rachel intends to buy some toiletries for her five brothers as a present. She thinks that Rex would like aftershave, Tom talc, Magnus body spray and the twins, James and Henry, would like shower gel. They all have a liking for "Jasmine" brand. Her budget is a maximum of £10.00 and if she can spend less so much the better. She finds that the items are available in combination packs as well as individually. Prices are:

Aftershave	£2.99
Body spray	£1.99
Talc	£1.99
Shower gel	£1.49
Aftershave & talc	£4.49
Shower gel & body spray	£2.99
Body spray double pack with free talc	£3.95
Aftershave & shower gel	£3.49

What is the least amount of money Rachel must spend?

A £7.48

B £8.46

C £8.47

D £8.97

E £9.95

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16 (Q32) A survey claimed that in 1990 40% of the adult British population smoked. It claimed further that in 1991, after an anti-smoking campaign, only 30% of the adult British population smoked. It reported a 10% fall in the number of smokers. However, the campaign organisers, in their advertising, reported a 25% fall in the number of smokers.

The campaign organisers' advertisement reflects:

A the change in the number of smokers expressed as a percentage of the population in 1990.

B the change in the number of smokers expressed as a percentage of the population in 1991.

C the change in the number of smokers expressed as a percentage of smokers in 1990.

D the change in the number of smokers expressed as a percentage of smokers in 1991.

E the change in the size of the population expressed as a percentage of the population in 1990.

17 (Q36) The fuel consumption of Ted's new car is given in the manufacturer's data sheet as follows:

Speed	*50km/h*	*90km/h*	*110km/h*
Consumption l/100km	7.5	6.0	6.5

The cost of fuel is 80c per litre. Ted's journey to work consists of 20km on country roads normally at 90km/h, 40km on motorway at 110km/h and 20km in town at 50km/h.

How much does his drive to work cost him?

A $4.16
B $4.24
C $4.26
D $4.40
E $5.30

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18 (Q37) Maggie is preparing 300 information packs so that she can respond to enquiries for her holiday cottages. Each pack contains:

- A basic information sheet - this is printed on A4 sized paper on both sides so that it may be folded to form 4 pages.
- A sheet with prices on one side and a booking form on the other - two of these are printed on a sheet of A4 paper which is then cut in two.
- An A4 information sheet giving details of some local amenities - this is printed on one side only.
- A single A4 sheet colour brochure printed on both sides - this is provided free by the local Angling Association.

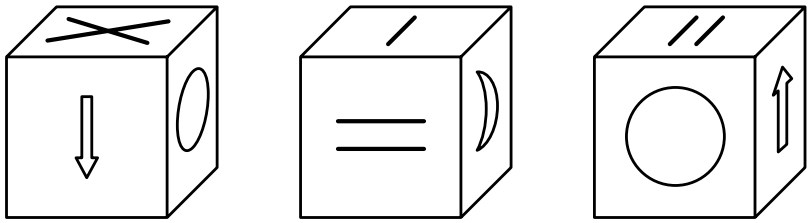
The local printer charges 2p per side of A4 for reproduction.

How much will her 300 packs cost her?

A £ 15
B £ 24
C £ 30
D £ 36
E £ 42

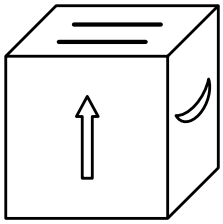
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19 (Q38) Three views of the same die with unusual symbols are shown below:

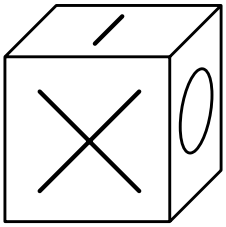


Which one of the following dice is **not** identical to the die above?

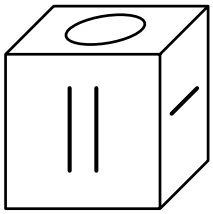
A



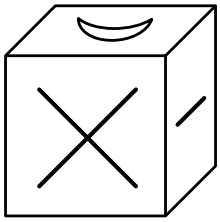
B



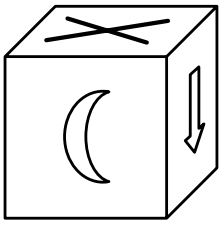
C



D



E



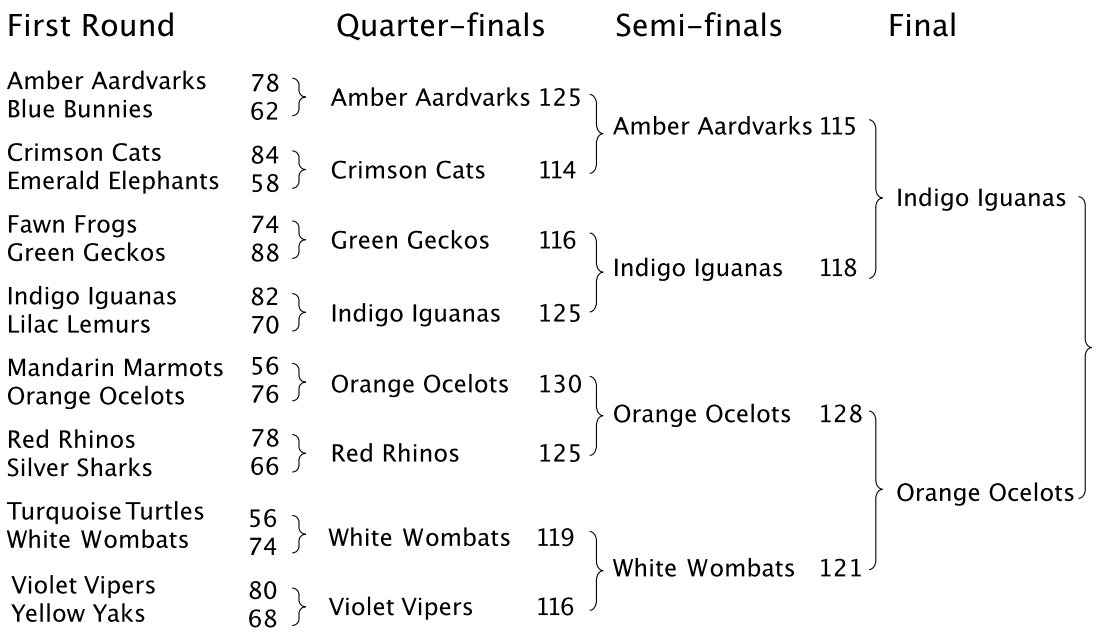
20 (Q42) The coffee machine in our office offers a choice of regular or decaffeinated coffee; black or with one of three sorts of milk (full, semi-skimmed, skimmed); and, for those who want sweetening, either one or two spoonfuls of sugar or one or two non-sugar sweetening tablets.

How many different kinds of coffee can be made?

A 11
B 24
C 32
D 40
E 48

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21 (Q43) This is the draw for this year's end of season knockout trophy (and the results so far) competed for by sixteen teams that take part in a quiz league:



Although teams are drawn against each other, all participating teams in each round answer the same set of 50 questions, worth 2 points each.

In the first round, each 'match' is won straightforwardly by the team with the higher number of points. This team carries half of these points through to the next round; then in subsequent rounds winners carry through half of their 'new' points from the previous round. For example, the Amber Aardvarks carried 39 points through to their quarter-final, then in the quarter-final itself they added 86, and so carried 43 through to their semi-final.

How many more points (at least) must the Indigo Iguanas score than the Orange Ocelots in the final to win the trophy?

A 4

B 6

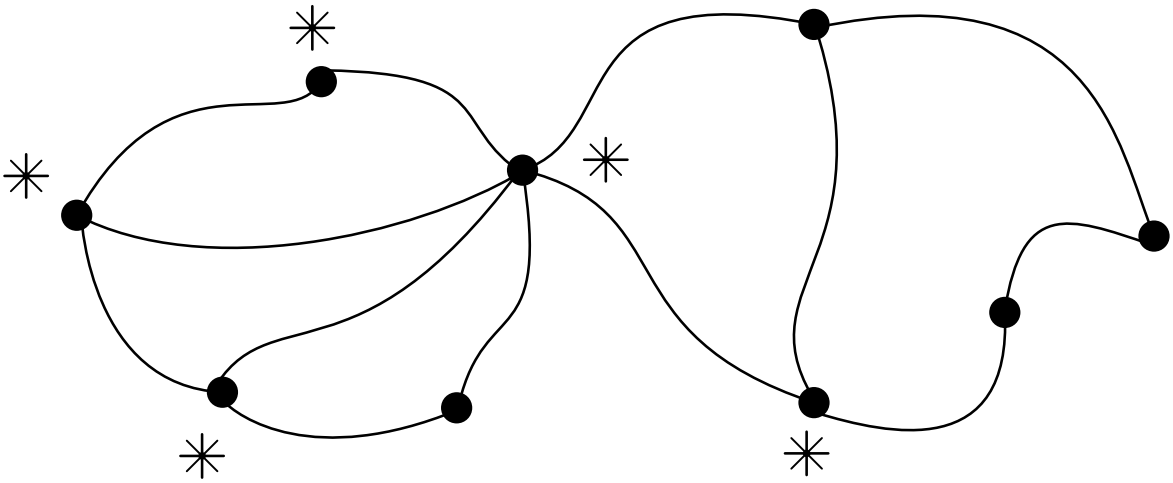
C 8

D 10

E 12

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22 (Q44) My local park has a network of paths linking various locations, as shown below. The stars on the map represent five of the park's landmarks: the fountain, lake, bandstand, statue and bowling green.



When I walk round the park, I often vary my routes and I do not always take the shortest route from one place to another. I have noticed that when I go from the statue to the fountain, I always pass the bandstand, but that going from the lake to the bowling green, I never pass the fountain unless I pass the bandstand twice.

How many of the stars on the diagram can be identified from the given information?

A 0
B 1
C 2
D 3
E 5

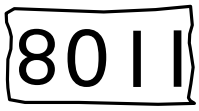
23 (Q48) The ferry from Harport to Selmer is at sea for 1 hour 15 minutes and at its berth at either end for 30 minutes. The hovercraft service between the same ports takes only 45 minutes and is berthed for 15 minutes before returning.

If both leave Harport at 08:00 what is the next time they will both be at the same port together?

A 09:45
B 12:45
C 14:45
D 15:00
E 18:00

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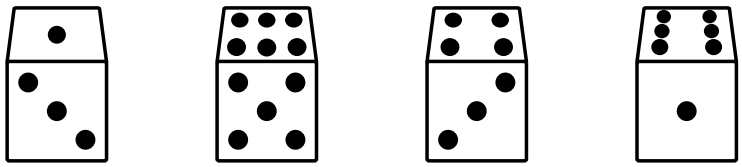
24 (Q49) All the telephone numbers in my village have six digits and begin with 58. I have accidentally torn up and thrown away a piece of paper on which I had written the telephone number of a friend who lives nearby. The only piece I can find is the scrap shown below:



How many different phone numbers in my village could be consistent with this extract?

A 2
B 10
C 11
D 12
E 22

25 (Q50) On a normal die, opposing faces add up to 7. Four views of the same die on which the numbers are arranged differently from a normal die are shown below. The unseen face has the number two on it.



How many spots are on the face opposite the two on this die?

A 1
B 3
C 4
D 5
E 6

END OF TEST

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